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A Self-Tuning Reinforcement Learning–Driven Isolation Forest Framework for Scalable and Efficient Anomaly Detection in Solar Power Plants

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Abstract
Anomaly detection plays a vital role in ensuring the stability and efficiency of the solar power production, where the accurate monitoring of the abnormality patterns can prevent failures and improve the performance. The proposed work introduced a new approach to anomaly detection through the integration of reinforcement learning (RL) and the Isolation Forest algorithm to improve the performance of anomaly detection in solar energy generation systems. This study aims to develop an adaptive and scalable solution to maximize the contamination rate of Isolation Forest model dynamically with Proximal Policy Optimization (PPO) based on a RL agent. The RL agent modulates the contamination rate of the Isolation Forest model through adaptive learning, enabling the system to adjust and evolving data patterns without manual intervention. It includes four main steps: (1) preprocessing the data (2) training the RL agent to maximize the contamination rate to best detect anomalies, (3) evaluating the performance of the RL-tuned Isolation Forest model (4) examining the detected anomalies. This technique is novel in that the adaptation of a key parameter (contamination rate) of the Isolation Forest algorithm (usually set manually or by heuristic), is performed via RL to make the system more adaptable and adaptive. The proposed model outperforms other mainstream anomaly detection models, e.g., neural networks and mainstream machine learning methodologies, achieving accuracy, recall, and F1 score of 0.94, 0.99, and 0.97, respectively. The results support the ability of the model to identify anomalies significantly below a balanced performance on several measures. The results of the current research point to the possibilities of applying reinforcement learning to the optimization of anomaly detection algorithms with an available scalable approach that can adapt to evolve and react according to the emergent and advanced trends in the operation of solar energy production and even in the overall renewable energy control systems.

Keywords Deep learning · Solar power · Reinforcement learning · Isolation forest · Anomaly detection

1 Introduction

Due to its availability, scalability and sustainability, solar energy is one of the most promising, and most popular renewable sources of energy, with large-scale energy production facilities supplementing or substituting fossil fuel sources, as well as providing on-site energy independence to residential and commercial applications [1].

Solar energy has a number of advantages, such as being virtually an endless source of energy. In addition to its environmental benefits, solar power can reduce residential electricity expenditure over time by reducing reliance on the old grid syatems. It is also a good alternative energy source in remote areas or poorly developed areas where access to the grid is either limited or unavailable [2]. Other than these advantages, however, there are also a series of challenges with the inclusion of solar energy to existing power grids, with the most typical challenge being the inherent changeability of the renewable sources. The differences in weather patterns, geographical positioning, and seasonal conditions create the inequality in the production of solar power. In order to reduce these irregularities and cseefficient operation of the energy system, the advanced monitoring and forecast systems are required [3]. Anomaly detection is an imperative aspect of management of solar power generation. Deviations in energy production can be the result of a faulty equipment, malfunction of the sensors or any unexpected environmental alterations. Such deviations properly identified are needed to create a secure system, avoid energy wastage, and enhance the overall performance of solar installations. There exists a popular notion that the traditional methods of anomaly detection take fixed detection thresholds or heuristics [4]. These mechanisms are fairly simple but they fail to work when working with nonlinear and dynamic solar energy data, so they classify system errors incorrectly or in a timely manner. To overcome these shortcomings, the application of ML in solar power has been widely adopted over the last few years, and the outcomes were promising in terms of the processing and analysis of the huge amount of data generated by a solar energy system [5, 6]. These techniques are expert at extracting the data patterns to predict and detect anomalies more effectively. All their strengths considered, most ML models are painstakingly sensitive to hyperparameter adjustment and insensitive to rapid-changing conditions, which confines them to real-time application. This paper suggests a new methodology by integrating Reinforcement Learning (RL) with detection of anomalies in solar power generation process.

In contrast to typical ML models [7, 8], RL learns to optimize actions by trial and error, and therefore it is very well suited for dynamic environments such as solar energy anomaly detection where the distributions of data are changing. At the core of the outlined framework which is a hybrid approach using PPO-based RL and the Isolation Forest algorithm. Isolation Forest is a general-purpose unsupervised anomaly detection method that identifies anomalies by iteratively partitioning data points using a recursive and binary tree algorithm. Although performing well, the algorithm's accuracy greatly depends upon choosing a right contamination rate—an input parameter representing the percentage of anomalies in a dataset. The determination of the optimal rate of contamination can be quite difficult and may involve domain knowledge or trial-and-error.

The suggested framework removes this constraint by using an RL agent to learn how to maximize the contamination rate. In this case, the reinforcement learning (RL) agent works on the basis of an anomaly detector, and rewards are attributed as the detection accuracy and precision, and the rate of contamination is adjusted to achieve the highest accumulation of the reward. The combination of the RL with the Isolation Forest algorithm features making the system flexible, as it can be used successfully in different data sets and solar power generation contexts.

The article aims at achieving the following objectives and contributions:

The purposes of the study are:

i. Dynamic anomaly detection: To design a scalable system based on reinforcement learning, which detects anomalies in solar power installations in the most efficient way possible through dynamic tuning of the most important parameters.
ii. Greater forecasting: To use the adaptive capacity of reinforcement learning to enhance precision and reliability in solar power forecasting and anomaly detection.
iii. Reality-on-the-field applicability: To guarantee that the offered framework is resistant to a wide range of geographical and climatic as well as operating conditions, and thus serves global implementation in the renewable energy industry.
iv. Grid stability and operational efficiency: It is use to detect the anomalies in the solar generation timely and accurately.

The paper suggests a new RL-based approach that integrates the flexibility of the reinforcements learning approach with the discriminative capabilities of the Isolation Forest algorithm to fulfill the particular paradigm that is needed in the case of solar power anomaly detection. The proposed approach can lead to a more sustainable and resistant energy future by improving the credibility of solar energy systems integration and performance on the global scale, thereby becoming more reliable, scalable, and precise.

2 Related Work

Ahmed and Khalid [9] carried out a study to increase the knowledge base of forecasting models for renewable power generation. Their survey, consistent with the findings of the power industry, indicated trends and made predictions for eventual developments. In solar and wind energy prediction, Zendehboudi et al. [10] discovered that models based on the support vector machine (SVM) were better than other techniques. Also, hybrid SVM models performed better with respect to the accuracy of forecast compared to non-hybrid SVM models.

Authurs [11] explored algorithms that predict renewable energy with deep learning which is based upon stacked autoencoders, deep belief networks, recurrent neural networks. Similarly in [12] proposed ANN based approach to estimate energy and analyse reliability. Their research accounted for different renewable energy resources, such as solar, hydraulic, and wind. Multiple examples were created in order to emphasize the power of ANN in energy forecasting, reliability analysis, and other uses. Researchers in [13] utilized advanced algorithms based upon ML for solar energy forecasting. The research utilized data gathered in Errachidia province from 2016 to 2018. Findings showed that GRU and RNN performed marginally better than GRU, especially in detecting long dependencies in time-series data. The RNN model recorded a Mean Absolute Error (MAE) of 1.83, a Mean Square Error (MSE) of 8.53, and a Root Mean Square Error (RMSE) of 2.92.

The authors proposed stacked ensemble (DSE -XGB) in [14] an integration of artificial neural networks and long short-term memory networks as base predictors with the aim to maximize their forecast reliability in different scenarios. It was experimentally demonstrated that DSE-XGB performed better, and the coefficient of determination (R 2) increased by 10–12% compared to benchmark models.

The paper [15] proposed a predictive system which used various machine-learning methods such as a fuzzy inference system (FIS), multi-layer perceptron feed-forward neural network (MLFFNN) a support votes regression (SVR), and an adaptive neuro-fuzzy inference system (ANFIS). Based on practical datasets of Iran, the research has found that the maximum accuracies were found to be with SVR and MLFFNN with corresponding R^2 values of 0.9999 and 0.9795 respectively.

Both, deep-learning algorithms including artificial neural networks (ANN), convolutional neural networks (CNN), and recurrent neural networks (RNN), and traditional machine-learning models such as, polynomial regression, SVR, and random forest were evaluated in [16] to forecast solar-radiation over four locations in Nigeria in a period of twelve years. The deep-learning models performed better as they gave an R^2 of 0.9546 and root-mean-square error/mean-absolute-error of 82.22 Wm -2 and 36.52 Wm -2, respectively.In [17] researchers evaluated four forecasting techniques namely KNN, LR, ANN, and SVM. Although all produced satisfactory predictions, ANN achieved the highest accuracy with RMSE of 86.446 and MAE of 8.409, outperforming KNN (92.857 / 8.8279), LR (94.5 / 8.96), and SVM (93.644/9.96).

In [18], time-series forecasting techniques, were applied to PV generation data from South Korea. Among them, LSTM achieved the best performance with an R^2 of 0.943 and an average Mean Absolute Percentage Error (MAPE) of 5.79.

Study [19] compared SVM and GBR for solar forecasting using Kaggle's Oklahoma dataset. Feature selection methods such as Linear Correlation, ReliefF, and a novel local information-based technique were evaluated. Results showed that nonlinear approaches consistently reduced prediction errors compared to linear ones.

Authors of [20] focused on integrating renewable energy into the grid based on site-specific ML forecasting models constructed using National Weather Service (NWS) data. Both SVM and LR were tested and SVM was more accurate and less erroneous compared to LR.

Reference [21] presented an enhanced RBFNN model optimized with K-means based algorithms and prediction of solar energy in Palestine (Jericho city). The improved RBFNN significantly outperformed standard RBFNN and MLP models, achieving its best RMSE of 0.004994 with only seven hidden neurons.

In [22], a hybrid deep learning framework was proposed, combining clustering methods, CNN, LSTM, and an attention mechanism supported by a wireless sensor network. Historical solar data from Shaoxing, China was used, and the hybrid model achieved an RMSE of 2.04, demonstrating improved accuracy over individual models.

Study [23] developed an LSTM-based RNN to predict monthly PV output in South Korea, with the aim of estimating production potential at new sites. Tested on unseen datasets, the model achieved an RMSE of 7.416% and a MAPE of 10.805%.

In [24], a hybrid Ensemble Average (EA) approach was proposed, consisting of a combination of machine learning models using on a 1 MW solar park and achieving lowest average absolute error (MAE) of 0.1295.

Lastly [25], introduced CNNLSTM hybrid model to predict stable solar energy; in this model, the convolutional neural network was used to learn meteorological relationships, and the long shortterm memory network to learn the temporal relationships. The system showed a mean absolute percentage error (MAPE) of 4.58 on clear days and 7.06 on cloudy days using data of Busan, Korea, which confirms its strength in different climatic conditions.

[26] In the recent past, statistical, machine-learning, and deep-learning procedures have been applied to the detection of anomalies in Internet of Things (IoT) and energy-related systems in large quantity. According to the survey literature, the real-world situations of IoT will be marked by the systemic imbalance of data, the lack of exhaustive ground-truth labels, and resource limitations, which are in turn encouraging the implementation of either unsupervised or hybrid anomaly-detection methods to be adopted in the actual practice [27]. Isolation Forest, Random Forest, artificial neural networks, long short-term memory networks and deep autoencoders are also learning based methods that have been effective in detecting anomalies in smart homes, sensor networks, and cyber-physical systems with improved detection accuracy under complicated conditions of operation.

The recent developments in the power systems and energy analytics have been based more on the data-driven learning methods, which assist in monitoring, predicting, and analysis based on faults. A fault diagnosis and power quality analysis implementation based on deep learning has shown good performances with regards to the ability to analyze the time-frequency features utilising lightweight convolutional neural networks to filter out the complex and noisy operating conditions [28, 29]. Hybrid deep learning and optimization models have also been successfully used in short-term load forecasts and transformer oil temperature forecasts in power system forecasting systems; they have successfully predicted nonlinear and non-stationary trends in energy data [30, 31]. Other such learning techniques applied to renewable energy systems such as wind speed prediction and battery state-of-charge prediction have been also applied have variable environmental conditions where robustness under changing environmental conditions is paramount [32, 33]. A number of studies also indicate that ground-truth fault labels are not often accessible in practical energy systems, which causes a switch to using unsupervised learning protocols and relative performance metrics instead of using strict supervised performance evaluation [34]. Also, metaheuristic optimization methods have extensively been studied to enhance convergence, flexibility, and resilience in complicated engineering and energy administration issues [35, 36]. Together, these studies encourage the design of adaptive and self-tuning learning systems that may be useful towards effective functioning in real-world conditions with limited labelling.

3 Proposed Framework

The proposed approach is used to reduce the increasing global energy demand at the same time as being able to reduce the harmful environmental effects linked to the use of traditional energy sources. The solar power generation systems though naturally sustainable are yet prone to myriads of anomalies that significantly influence the generation capacities as well as performance indicators. These aberrations can occur due to varying causes like machinery breakdowns, weather, system wear and immediate alterations of the environment. The timely detection of such anomalies in real time is also the key to maintaining the continuous and best operation of the solar power systems hence avoiding the downtime and limiting costs to maintenance.

Conventional anomaly detection schemes such as statistical, rule-based or machine-learning-based methodologies are not so much effective when used for solar power system. The nature of these methodologies is fixed, which uses fixed parameters, and thus, they are not sensitive to the changing behavior of solar energy production. The factors that include diurnal cycle, seasonal variation and weather parameters have a significant effect on the power generation and the old models are not responsive enough to this effect. Furthermore, there are significant issues in the process of balancing anomaly detection and false positive reduction because false alarms of anomalies can stimulate unnecessary maintenance operations or the shutdown of a system. The current research paper contains a novel approach to a combination of Reinforcement Learning (RL) and the Isolation Forest algorithm. The dynamic adaptation of parameters, including the contamination rate, by an RL maximizer based on feedback of solar power is novel. With the highly successful RL algorithm Proximal Policy Optimization (PPO), the agent learns through trial and error to enhance performance, refining anomaly detection incrementally.

RL with Isolation Forest enhances solar power anomaly detection through the ability to keep the model flexible for learning from changing data distributions of weather and system behaviour. Such models are less likely to overlook newly emerging anomalies compared to static methods. Additionally, varying the contamination rate in the Isolation Forest enhances model sensitivity, which is high in precision and recall. This is important in solar power systems where false alarms and overlooked anomalies can cause inefficiency in operations. This model is scalable to process large data sets in solar power systems and can be retrained from new data and is therefore appropriate for deployment over long periods in real-world situations. The different module of proposed framework is given below.

3.1 Module of Proposed Framework

The architecture of the RL-guided Anomaly Detection using Isolation Forest is composed of four substantial phases in the architecture: data pre-processing, configuration of the RL, RL agent training, testing and anomaly investigation. The key element of the presented model is shown in Fig. 1.

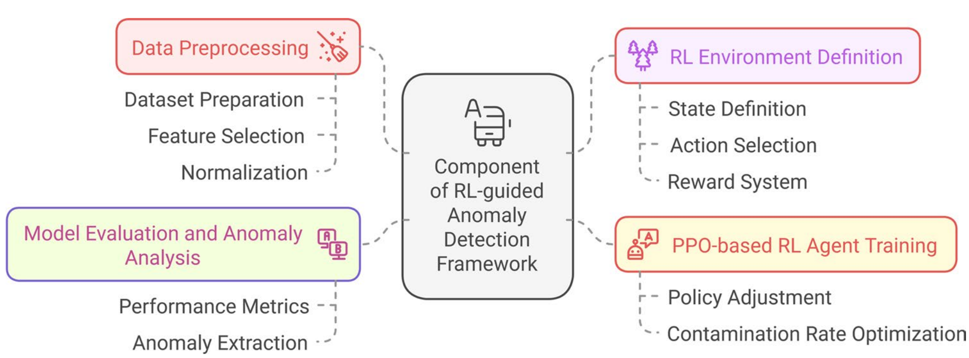

Fig. 1 Component of proposed framework

3.1.1 Data Preprocessing

The preprocessing of dataset is the first step of the framework where data is cleansed and prepared for analysis before use in machine learning models. The D dataset includes significant features such as Day of Year, Average Temperature (Day), and Power Generated. These characteristics extract the most critical information that determines solar power generation. Data cleaning and feature selection are done to delete any missing or irrelevant data points, and synthetic labels, Y, are created for training. Training labels are signifying whether the data point is normal (1) or abnormal (– 1) as shown below:

dataset denoted as:

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N} \tag{1}$$

where each data point $x_i \in \mathbb{R}^3$ is a feature vector:

$$x_i = \{\mathrm{DOY}_i, T_{\mathrm{avg},i}, P_{\mathrm{gen},i}\} \tag{2}$$

and $y_i \in \{1, -1\}$ is the label indicating:

$$y_i = \begin{cases} 1, & \text{if normal} \\ -1, & \text{if anomaly} \end{cases} \tag{3}$$

Handling Missing Values: For any missing value in feature f, impute as:

$$f_i = \frac{1}{N_f} \sum_{j=1}^{N_f} f_j, \tag{4}$$

where f_i is missing and N_f is the number of non-missing entries.

Synthetic labels are generated solely for RL agent training and performance evaluation. The Isolation Forest algorithm operates in an unsupervised manner and does not require these labels during its training or anomaly detection process.

Feature Normalization: To ensure all features contribute equally to model training, standard scaling is applied to each feature:

$$\acute{f}_i = \frac{f_i - \mu_f}{\sigma_f} \tag{5}$$

where μ_f is the mean of feature f, σ_f is the standard deviation of feature f, $\acute{f}_i$ is the normalized feature value.

Final Preprocessed Dataset:

$$\mathcal{D}_{\mathrm{norm}} = \left\{\left(\acute{x}_i, y_i\right)\right\}_{i=1}^{N}, \acute{x}_i = \left\{\mathrm{D}\,\acute{\mathrm{O}}\,\mathrm{Y}_i, \acute{T}_{\mathrm{avg},i}, \acute{P}_{\mathrm{gen},i}\right\} \tag{6}$$

This normalized dataset is then used as input to the RL-environment for adaptive anomaly detection. The dataset is normalized with standard scaling. This process makes all features have the same scale and avoids any feature from dominating the learning process because of varying magnitudes.

3.1.2 Reinforcement Learning Environment

The RL environment is where the PPO agent interacts with the data to learn optimal actions (i.e., the contamination rate) that increase the model's efficiency. In this environment, the state is defined as the current data point X[i], which consists of the selected features (Day of Year, Average Temperature, Power Generated) and agent's action corresponds to selecting a contamination rate. The contamination rate regulates the proportion of data points that will be classified as anomalies during model training. The reward signal in this environment is designed to encourage accurate anomaly classification. If the model predicts the label correctly (i.e., matches the true label Y[i], the agent receives a reward of + 1, otherwise, it gets a reward of -1 as shown below:

At each time step t, the agent observes a state:

$$s_t = \{\mathrm{DOY}_t, T_{\mathrm{avg}\ ,t}, P_{\mathrm{gen}\ ,t}\} \tag{7}$$

The action taken by the agent is the selection of a contamination rate:

$$a_t = c_t \in [0.01{,}0.1] \tag{8}$$

The search space of contamination-rate [0.01, 0.1] was chosen with references to the specific nature of the solar power systems, as well as the demands of model stability. By the nature of how operational solar-power generation is structured, anomalies, such as sensor faults, inverter failures, shading effects and communication errors are actually very rare in nature, and make up a very small portion of the overall operation data. Therefore, by imposing the rate of contamination on a small range, realistic anomaly densities are reflected and excessively large abnormal behavior is prevented. The contamination values below 0.01 led to under-detection of subtle but operationally significant anomalies, while values above 0.1 resulted in excessive false positives and unstable reward signals during reinforcement learning. The suggested framework is not based on the predetermined contamination value. The given interval simply defines the action space that the agent of reinforcement-learning may act in (Eq. (8)). The desirable contamination rate c is adaptively learned using rewards based on the maximization (Eqs. (9) and (10)) and thus, allows the model to regulate its sensitivity to changes in the data distributions. Such limited but flexible formulation ensures vigor, consistent learning and viable practicality in real-life situations of solar-power anomaly-detection.

The Isolation Forest model uses this c_t to classify the data point X_t, and based on the predicted label $\acute{Y}_t$ and the true label Y_t, the reward function is defined as:

$$R\left(s_t, a_t\right) = \begin{cases} +1, & \text{if } \acute{Y}_t = Y_t \\ -1, & \text{otherwise} \end{cases} \tag{9}$$

The environment transitions to the next state s_{t+1}, and this process continues until a termination condition is met.The agent moves on to the next data point, and the operation is repeated until the entire data set has been processed or until a stopping point is reached. The objective of an RL agent is to maximize overall rewards by determining the best contamination rate for each portion of the data set such that anomalies can be detected properly over time. This capability of adjusting the contamination rate dynamically as per the data is a considerable strength over existing static anomaly detection models. The motivations behind using reinforcement learning in the paper are to model the contamination rate selection as a successive decision making problem as opposed to aspiration or heuristic parameter update. The given Proximal Policy Optimization (PPO)–based agent, in contrast to the rule-based or online optimization approaches, is conditioned with the help of accumulated reward feedback, thus, enabling the model to adjust a conditioned policy to changing solar power data characteristics.

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3.1.3 PPO-Based RL Agent Training

The RL agent is seeded with policy parameters that are randomly generated, and these are used as an initial point of departure, and it is trained on the Proximal Policy Optimization, better known as PPO as shown below:

The RL agent maintains a parameterized policy $\pi_\theta\left(a_t \mid s_t\right)$, where θ denotes the policy parameters. The policy is updated using the PPO clipped objective:

$$L\left(\theta\right)=\mathbb{E}_t\left[\min\left(r_t\left(\theta\right)A_t,\operatorname{clip}\left(r_t\left(\theta\right),1-\epsilon,1+\epsilon\right)A_t\right)\right] \tag{10}$$

where $r_t\left(\theta\right)=\frac{\pi_\theta\left(a_t|s_t\right)}{\pi_{\theta_{\text{old}}}\left(a_t|s_t\right)}$ is the probability ratio, A_t is the advantage estimate at time t, $\in$ is a small constant (typically 0.1 or 0.2) to control the update size.

The agent iteratively updates θ to maximize the expected cumulative reward until a convergence criterion is met (e.g., total time steps T or reward threshold).

The convergence of Proximal Policy Optimization (PPO) agent will be guided by vigorously defined stopping conditions to ensure acquisition of robust and efficient learning processes. Convergence of PPO training is considered to have been achieved when either of the details mentioned below satisfies. This is first attended to in order to provide a limit on the complexity of calculations by termination of training when the upper limit of interaction time steps (T) has been surpassed. Second, an early-stopping policy based on the performance of the policy is in place: should the moving average of cumulative episode rewards demonstrate no statistically significant growth over an improvement beyond some trivial threshold, the learning is stopped, which is denoted by: ϵ .This two-way convergence strategy ensures that the agent of reinforcement-learning achieves an effective policy in the selection of the contamination rate, and reduces to the same time the risks of over-training and oscillatory behaviour. The proposed framework optimises the time-based and reward-based convergence conditions, which makes it reliable and reproducible in the dynamic anomaly-detection environment.

The PPO is one of the best reinforcement learning algorithms, in large part because it can trade off exploration of new policies, versus exploitation of known successful policies. Such balance is needed because it helps the agent learn optimally, thereby minimizing the possibility of getting stuck in suboptimal solutions that cannot lead to the best possible results. The mechanism through which the PPO algorithm operates is to update the policy by modifying its parameters based on environmental feedback, using a surrogate objective that is carefully crafted to allow such policy parameter modifications. For each training episode, the agent interacts actively with the environment by taking decisions out of available actions, which are associated with various levels of contamination, and the agent is rewarded that is representative of the quality of its anomaly detection performance. Over time and through repeated interactions, the agent will be able to optimize the rate of contamination more and more, leading eventually to an improvement in the performance of the Isolation Forest model to a level where it can detect with better accuracy anomalies that are embedded in the solar power data. Training for the agent is continuous until a specified number of time steps, known as T, has been attained, or otherwise until optimal training solution is achieved.

The proposed framework that is based on the end-to-end architecture illustrated by Fig. 1, shows the flow of interaction between the data pre-processing, the PPO-based reinforcement learning agent and the Isolation Forest model, thus showing how feedback based on the performance of the anomaly detection is continually being exploited to tune the contamination rate.

3.1.4 Model Evaluation and Anomaly Analysis

Following successful completion of the training process by RL agent, the following process involves training an Isolation Forest model on the basis of the best contamination rate, represented as c∗, that has been identified by the RL agent. The model after tanning is subsequently used to forecast the dataset to be either normal, assigned as a score of 1, or abnormal, assigned as a score of 0When calculating the efficiency of anomaly detection, a number of performance metrics are calculated, which entails a holistic evaluation that separates data points and labeled outliers. Also, the relative performance is determined in the standard operating environments. Anomalies are the points characterized by outliers by a trained Isolation Forest model. The further examination of the observed anomalies is expected to clarify their features and reasons behind their appearance, creating the knowledge that could be used to shape maintenance measures and aid the processes of making operational decisions.

3.2 Proposed Algorithm

This algorithm description is given to present the proposed Isolation Forest framework guided by reinforcement learning in a clear step by step procedural form. Although the inner building blocks (data preprocessing, PPO-based learning, and anomaly detection) have been mentioned above, here they are united into one full execution cycle that offers the focus on how the contamination rate should be optimised dynamically in the course of execution. It is focused on the clarity of implementation and reproducibility as opposed to repeating previously presented explanations.The method it uses to detect anomalies in solar power, which can be denoted as the framework of RL-Guided Anomaly Detection, combines the scientific fields of Reinforcement Learning (RL) and the Isolation Forest algorithm in a new and efficient way. With this hybrid formulation, real time optimization of important model parameters is realized whilst simultaneously detecting dynamic anomalies. Repeatedly, the manipulation of the RL and the Isolation Forest strengths allows the system to approach the variable nature of the solar energy data. The Fig. 2 below provides a comprehensive explanation and graphical illustration of the proposed framework and highlights the repetitive learning process, clearly presenting the sequence of state observation, action selection (contamination rate), rewards calculation, and policy adjustments replacing each other in a dismissal way. It provides clear understanding of how reinforcement learning is embodied into the anomaly detection and the newness of adaptive parameter tuning.

In the proposed framework, the reinforcement learning agent interacts with the Isolation Forest in an episodic manner, where the contamination rate is updated at each decision step based on the current data state. The Isolation Forest is retrained using the selected contamination rate, and policy updates are guided by reward feedback until convergence criteria based on time steps and reward stabilization are met. The general scheme is based on an iterative learning scheme whereby a reinforcement learning agent is trained to vary the rate of contamination of the Isolation Forest which is a simple parameter that dictates the sensitivity of the model to abnormal measurements. In order to maintain the stability of anomaly detection, by varying this rate of contamination dynamically, it is hoped that the stability under conditions of varying underlying data distribution due to solar system measurements can be maintained as that data distribution can vary over time. The algorithm begins by initially loading

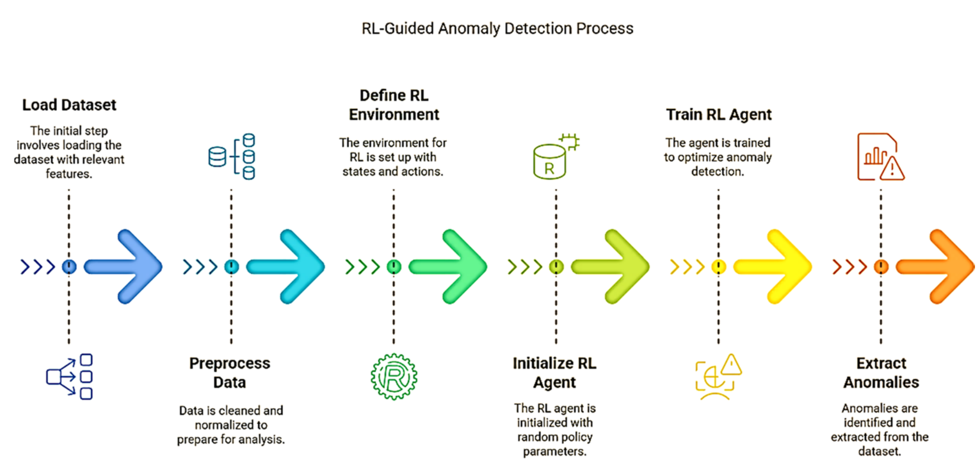

Fig. 2 RL-guided anomaly detection process

Algorithm RL-Guided Anomaly Detection using Isolation Forest in solar power generation

Input:

1. Dataset D with features F = {Day of Year, Average Temperature (Day), Power Generated}
2. Ground truth labels Y (1 for normal data, -1 for anomalies)
3. Contamination rate range c ∈ [0.01, 0.1]
4. RL training parameters: Maximum time steps T, stopping threshold ε.

Initialization:

1. Data Preprocessing:

Load dataset D.

Select relevant features F and remove missing values using Equation (4).

Normalize all features using Equation (5).

Construct final preprocessed dataset D_{norm} as described in Equation (6).

Generate synthetic labels as per Equation (3).

2. Define RL Environment:

Define State S_t using Equation (7).

Define Action $a_t = c_t \in [0.01, 0.1]$ using Equation (8).

Employ c_t to fit the Isolation Forest model .

Predict the label $\hat{Y}_t$, compare with Y_t , and compute reward using Equation (9).

Transition to the next state s_{t+1}. Terminate when end of dataset is reached.

3. Initialize PPO-Based RL Agent πθ(a|s):

Initialize PPO-based RL agent $\pi_\theta(a_t \mid s_t)$ with random policy parameters θ.

4: Train the RL Agent

For t = 1 to T:

4.1 Reset the environment E to start from the first data point X[0].

4.2 While the episode is not done:

Select action $a_t \sim \pi\theta(s_t)$.

Set contamination rate $c_t = a_t$.

Employ c_t to train Isolation Forest M

Predict label $\hat{y}_t$

Compute reward $R(s_t, a_t)$ using Equation (9).

Transition to next state s_{t+1}

Update policy $\pi_\theta(a_t \mid s_t)$ with objective defined in Equation (10).

5: Use the Trained RL Policy

Extract the optimal contamination rate $c^* = \pi_\theta(s)$.

Employ c^* to train the Isolation Forest M

Predict labels ŷ for all data points in X.

6: Evaluate Performance

Convert labels Y[i] = -1 → 0 (anomaly), Y[i] = 1 → 1 (normal).

Compute evaluation metrics: Accuracy, Precision, Recall, F1 Score, ROC-AUC.

If metrics exceed threshold ε, proceed to Step 7. Otherwise, return to Step 4.

7 Extract anomalies:

$A = \{X[i] \mid \hat{y}[i] = 0\}$.

Report the number of anomalies |A| and their details.

Output:

1. Trained Isolation Forest model M
2. Optimal contamination rate c*
3. Performance metrics
4. Detected anomalies A.

the dataset D, which is a complete set of data expressly relating to the generation of solar power. The dataset contains significant information like the Day of the Year, Average Temperature, and Power Generated. Features of the dataset are preprocessed in anticipation of analysis in order to remove any missing values and scale the data appropriately. This preprocessing phase ensures that the model is being provided clean and correctly scaled inputs necessary for correct functioning. Ground truth labels Y are subsequently created, which play a very critical role by showing whether the operation is normal (label 1) or abnormal (label − 1). Once the dataset is prepared accordingly in input features and labels, the next step is preparing the reinforcement learning (RL) environment. The state in the provided described setting is the location of data X[i] at the moment of time, while the action is a floating value because the contamination rate (c). This contamination rate is a vital parameter utilized in setting the sensitivity in anomaly detection using Isolation Forest model. The agent gets rewarded in terms of prediction accuracy: +1 as reward when the predicted label equals the actual label, while vice versa, the agent gets punished with − 1 in case there is any mismatch. The agent moves to the next data point X[i + 1] and does this for some time until it has passed through the whole data set or a stop criterion is reached. In this training, the RL agent begins with its policy parameters randomly initialized and undergoes learning through Proximal Policy Optimization (PPO) across a series of episodes, enhancing its predictability in the process.In each and every episode which is presently being experienced, the agent experiences an example of a specific action, with the specific focus being placed on the rate of contamination. It then makes use of this rate of contamination in such a manner so as to train the Isolation Forest model efficiently. After training the model, it is run to predict that each data point belongs to class normal or is an outlier. The mechanism of expected reward is such that the accumulation of rewards is dependent on whether the predictions are close to the ground truth labels or not. The RL agent is subjected to optimization steps that try to maximize cumulative reward in the long term, thereby enabling it to find its best contamination rate. After completion of the training phase, the best contamination rate $c*$ that is represented by the RL policy is obtained. Once derivation has been completed, the Isolation Forest model is then retrained on this maximum value thus achieved. Upon achieving this retraining, the model is then utilized to provide predictions for the whole dataset, and this makes it possible to tag every individual piece of data within that set. The labels are tested against key performance metrics. If the performance metrics are above a specified threshold ∈, which represents good model performance, the process is complete. Otherwise, the training loop is iterated to further improve the model. Anomalies are finally identified by finding the data points where the predicted label shows an anomaly (i.e., label 0). The algorithm provides the count of anomalies and their information, useful into the functioning strength of the solar power system. The dynamic and adaptive quality of the RL-guided method enables the algorithm to customize the contamination rate, being more flexible and accurate in identifying anomalies under different conditions. This makes the proposed technique especially suitable for practical use in solar power generation, where distributions of data may change over time as a result of factors such as weather, system degradation, or other environmental factors.

The adaptive and dynamic nature of the RL-guided method enables the algorithm to learn to optimize the contamination rate, thus being more flexible and accurate in identifying anomalies under evolving conditions. This renders the proposed approach highly appropriate for uses in solar power generation, wherever data distributions may change over time due to weather, system degradation, or other environmental factors.

3.3 Computational Complexity Analysis

This part performs a computational complexity study on the proposed reinforcement-learning-inspired isolation forest framework, and as such illustrates its practicality in efficient and scalable large-scale detection of solar power anomalies.The isolation forest algorithm has an average time significance of O(n log n), whereby n is the number of samples of data. The reason behind this complexity is the construction of several isolation trees each of which is built by random selection of features and recursive partitioning of the features. Since all the trees use subsampled data, the total cost of computation is manageable even with large data sets.The proximal policy optimization (PPO) is applied to implement the reinforcement learning component. The PPO agent works in a

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Table 1 Configuration of system

This table outlines the hardware and software components used for the proposed work in a structured manner.

Component	Details
Hardware	
Processor	Intel Core i7-9700 K (8 cores, 3.6 GHz base frequency)
RAM	16 GB
Storage	1 TB SSD
Software	
Operating System	Windows 10
Programming Language	Python 3.8
Libraries/Frameworks	TensorFlow 2.3.0 and Keras 2.4.3, Scikit-learn 0.24.2, Stable-Baselines3 1.2.0, Pandas 1.2.4 and NumPy 1.19.5, Matplotlib 3.4.3 and Seaborn 0.11.1

Table 2 Comparison of proposed model with existing models

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC	Anomalies Detected
RNN	0.93	0.93	1.00	0.97	0.50	0
SVM	0.89	0.93	0.96	0.94	0.50	37
KNN	0.90	0.93	0.96	0.95	0.51	37
ANN	0.93	0.93	1.00	0.97	0.50	0
CNN	0.93	0.93	1.00	0.97	0.50	0
Proposed	0.94	0.94	0.99	0.97	0.50	30

low-dimensional state space and maximizes one scalar control, which is the rate of contamination. The complexity of the policy- update, as a result, increases linearly with the training-steps number, which is denoted as 0(T). Since convergence within PPO is realized under a finite number of steps, the overhead cost of a reinforcement learning is reduced. The total computational power of the suggested framework can thus be indicated as

$$\mathrm{O(n\ log\ n + T)},$$

which proves that the algorithm can be efficiently scaled to larger sizes of dataset. Furthermore, model inference is still lightweight, which makes the framework appropriate in detecting anomalies. The suggested scheme has a desirable trade-off between the detection and the computing costs, hence making it possible to implement it practically in high-scale and continuously changing solar energy power systems.

4 Results and Discussion

4.1 Environment Setup

The proposed anomaly detection framework has reinforcement learning (RL) combined with the Isolation Forest algorithm. The environment setup is essential to ensure reproducibility and best model performance. Below is an elaborate description of hardware and software usage in the proposed work as shown in Table 1. Here is the hardware and software setup described in tabular form:

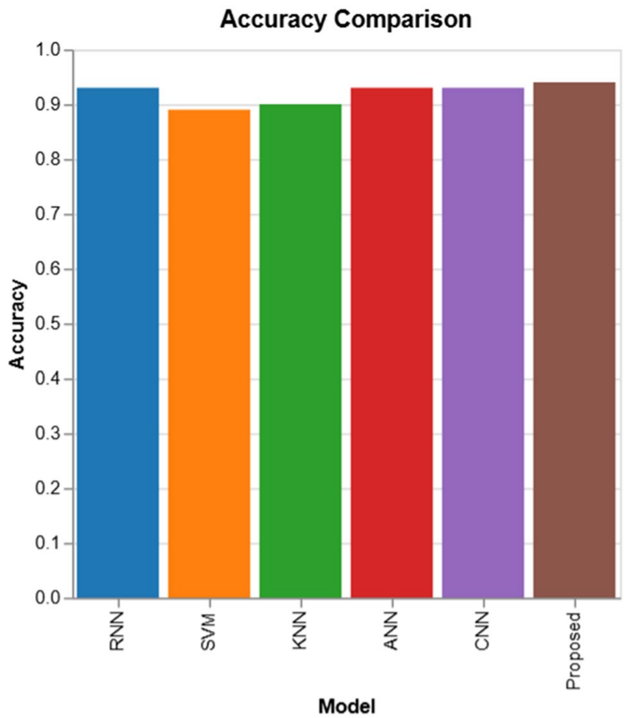


Fig. 3 Accuracy comparison

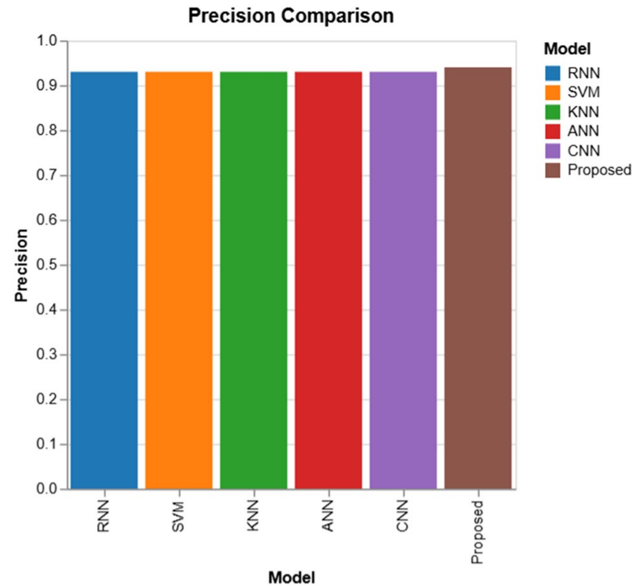


Fig. 4 Precision comparison

4.2 Results

The table below summarizes the performance of the models compared in this study including the proposed RL-guided Isolation Forest approach using performance metrics [37], as shown in Table 2:

The anomaly count is used as a relative indicator of model sensitivity rather than an absolute metric, since reliable ground truth anomaly labels are unavailable in real-world solar power datasets. There is a need to underline the fact that the reported Recall value of 1.0 of the RNN, ANN, and CNN models do not always indicate an effective anomaly detection operation. Due to the overly uneven structure of the dataset, such models are prone to stabilize to making all the instances observed as normal, which in turn results in the production of zero anomalies detected. In these cases, the Recall measure can seem artificially high, and should be viewed together with the measures of the number of anomalies and the measures of precision, but not as a stand-alone performance measure.

4.2.1 Accuracy

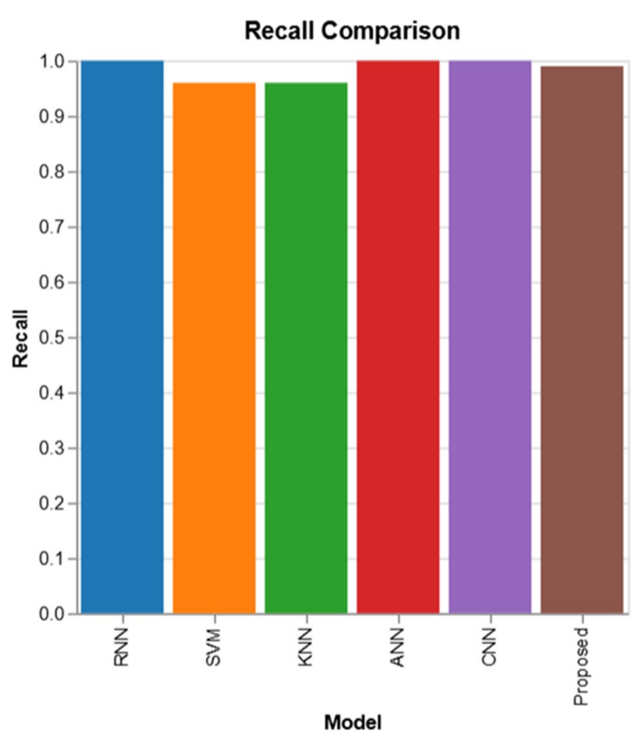
Perhaps the simplest, and certainly the most commonly used measure to gauge the overall performance of a classification model is accuracy. It is merely the proportion of accurate forecasts (normal and anomalous) to overall forecasts. For anomaly detection, high accuracy guarantees that the model will be able to distinguish normal data points from anomalous ones correctly. The Fig. 3 is a graphical comparison of the accuracy of each of the tested models, thus allowing a quick visual comparison of differences in relative models. Although Table 2 provides specific numerical numbers, it also is a precursor of better overall accuracy of the proposed model and further supports its strength in juxtaposition to traditional machine-learning and deep-learning models. The visual comparison resonates with the ability of the readers to determine the rankings in models without necessarily having to read the numerical tabulations The Highest Accuracy of 0.94 is achieved by the Proposed RL-guided Isolation

Forest model, indicating that it is very effective in accurately classifying normal and anomalous instances. This is better compared to existing models such as SVM and KNN, which have accuracies of 0.89 and 0.90, respectively. Interestingly, although models such as RNN, ANN, and CNN all have 0.93 accuracy, they fail to identify anomalies, which recommends that these models are overfitting on the normal data and are not good generalizers to anomalies. The Proposed Model is the most dependable one, having better accuracy while balancing the trade-offs with other measures such as precision and recall.

4.2.2 Precision

Precision is highly important in the detection of anomalies since it reflects directly on how well the model can prevent false positives, i.e., correctly classifying normal data points as anomalies. The high precision value indicates that one may be sure that the discovered anomalies are real anomalies rather than false positive samples misclassified as anomalies. Figure 4 highlights the differences in accuracy between models, which is a very relevant measure of relevance in the context of anomaly detection and reduces false positives. Although, the corresponding table records similar values of precision of the several models, the number graphically highlights the slight but significant gain achieved by the suggested methodology. In turn, this visual evidence reinforces the statement that RL-guided approach generates more reliable anomaly alerts. In this research, the Proposed Model has the highest precision of 0.94, which means that 94% of the predicted instances as anomalies are indeed true anomalies. The SVM and KNN models also have good performance, both having a precision of 0.93, which implies that they are equally effective in reducing false positives. The models, however, identify more anomalies (37), which can lead to a greater rate of false alarms. The models RNN, ANN, and CNN attain a marginally lower precision of 0.93, which indicates their ability to identify anomalies but not over-detecting instances as anomalies. Given that the Proposed Model attains the highest precision, it is an indication that it has a better balance with respect to detection, such that the detected anomalies are meaningful and the number of unnecessary alerts is kept low.

Fig. 5 Recall comparison



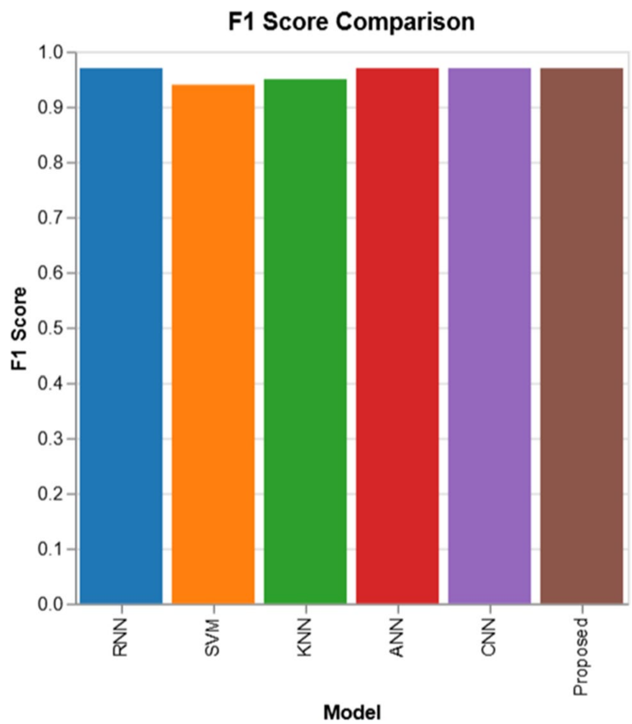


Fig. 6 F1 Score comparison

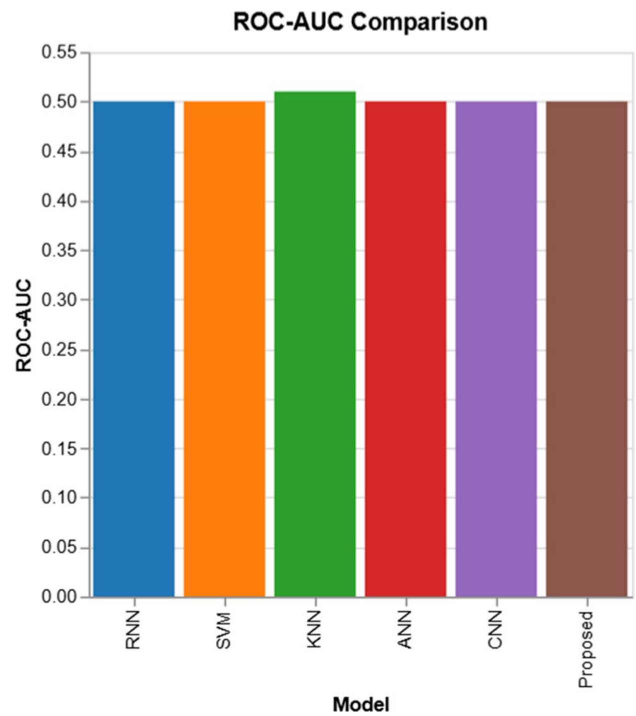


Fig. 7 Roc-AUC comparison

4.2.3 Recall

Important measurement in the sphere of anomaly detection is the recalls because it quantifies the ability of the model to identify all real anomalies in the data. Figure 5 graphically illustrates the performance of the entire models with regard to recall, which is a clear indication of the effectiveness of the anomaly detection. Although some models achieve perfect recall by numerical measures, the figure puts the number into perspective with the number of anomalies, thus showing that high recall does not consider a useful result in the context of low anomaly recognition. In such a way, visualization supports the balanced performance of the proposed model. The better the recall value, the better the model performs in detecting anomalies, which in scenarios where losing a single anomaly would have immense implications, e.g., solar power generation where anomalies that go undetected can cause the system to fail or be inefficient. All of the RNN, ANN, and CNN models have a 1.00 perfect recall, which means they are able to detect all instances of anomalies within the dataset. Yet, they only detect 0 anomalies, which means they might be overfitting the normal data and not classifying any data points as anomalies. In contrast, the Proposed Model attains a recall of 0.99, which, though less than ideal perfect recall, still demonstrates superb performance. The Proposed Model does pick up on 30 anomalies, which means that it is indeed able to find nearly all real anomalies in the data. In comparison to SVM and KNN (both of which identify 37 anomalies), the Proposed Model identifies slightly fewer anomalies, but still maintains high recall and does not mark unnecessary outliers, thus making it a better balanced solution.

4.2.4 F1 Score

There is absolutely no better way of measuring the effect of anomaly detection than the F1 score, which approximates the harmonic mean of precision and recall, i.e. it gives an optimum equilibrium between the false positive and false negative trade-off. Figure 6 accentuates the harmonic balance among precision and recall acquired by

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each of the models. Although the F1 -scores are tabulated in Table 2, this number makes it easy to appreciate the fact that the proposed model is always able to maintain a high F1 -score, and at the same time identify anomalies, which a number of deep-learning models cannot achieve. Based on this, the figure gives visual credence to the assertions of equivocal detection ability. A large F1 score means a better trade-off between false positive and false negative.

The Proposed Model has F1 of 0.97, indicating a very good balance between precision and recall which is very significant and important when the false positives and false negative (missed anomalies) should be reduced to minimal. Other models such as RNN, ANN, and CNN also have good F1 scores of 0.97, but since they fail to detect any anomalies, their F1 scores are not very useful in practice. SVM and KNN, though with slightly lower F1 scores of 0.94 and 0.95, respectively, indicate that they are still efficient in anomaly detection but at the cost of precision and recall due to more false positives. The Proposed Model performs better than these models in F1 score by having a strong detection ability while minimizing false positives and false negatives.

4.2.5 ROC-AUC

The ROC-AUC score is a significant indicator of the discriminative power that a prediction model possesses over a continuum of thresholds. Figure 7 presents the values of the ROC-AUC of each of the models, thus supporting the difficulties of working with extremely unbalanced datasets of anomalies detection. Despite the pronounced similarity between the numerical values, the visualization demonstrates that ROC-AUC should not be considered as the only measure of effectiveness in the case, which holds one to consider precision, recall, and anomaly numbers as more informative indicators as well. The score actually provides a quantitative assessment to which the model can perform in order to properly discriminate between those data points that are considered normal and those that are marked as abnormal. A good ROC-AUC score means that the model is highly capable of discriminating between these two distinct classes. In this study, all the models experimented with, i.e., the Proposed Model, had an ROC-AUC score of 0.50. This particular score means that the models do not have any notable discriminatory power, basically doing as well as random guess. This is a natural expectation in the case of datasets that are highly imbalanced, whereby there is an extremely dominant majority of ordinary instances against the extremely rare occurrence of anomalous data points. In spite of the low ROC-AUC score, however, It is also worth noting that the proposed framework performance is better than other models in terms of precision, recall, and F1 score. It means that the given model carries on being robust even when the imbalance in datasets has taken place. Of interest is the fact that low ROC-AUC values are typical of anomaly detection where the main aim of the models is to rightly detect anomalies and not to maximise the overall discriminative performance. The number of the anomalies revealed by the model can be considered as a correct measure of sensitivity of the model to outliers. The two models, Support Vector Machine (SVM) and K -Nearest Neighbour (KNN) models reveal the most number of anomalies (37) implying a highly sensitive nature to the possibility of an anomaly at the expense of a high failure positive rate. The proposed model identifies 30 deviations, which is a better tradeoff between specificity and sensitivity, hence, it does not lead to overfitting. The Neural Network models used have Recurrent Neural Network (RNN), Artificial Neural Network (ANN), and Convolutional Neural Network (CNN) and all of them show zero anomalies, meaning that either they do not recognize the anomalies or they overfit the normal data. The 30-anomaly detection which is exhibited by the proposed model, alongside, the good recall and the good precision, implies that the proposed model can be used in real world anomaly detection of systems like solar power generation where anomaly detection is the key to efficiency of the system. This demonstration represents an acceptable ratio between the identification of applicable anomalies, and the production of false alarms, which makes it more practical in these cases. The values of the ROC-AUC are close to 0.50 due to the strong imbalance of the data and the tendency of several predictive models to be biased towards the dominating normal class at all the thresholds. In these anomaly detecting settings, the ROC curve normally falls onto the diagonal hence the informativeness of the ROC-AUC measure is reduced. Consequently, the evaluation of the model performance is

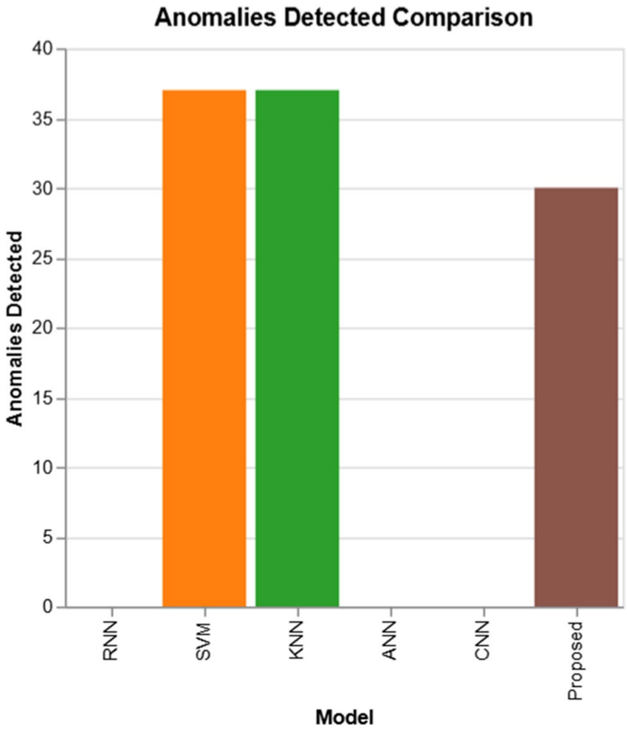

Fig. 8 Anomalies Detected comparison

overly dependent on the accuracy, recall, F1 -score, and the number of false positives, which is more representative of the realistic detection potential.

4.3 Discussion on Model Comparison

Comparison of the presented reinforcement-learning-guided Isolation Forest model to the already existing machine-learning models (RNN, SVM, KNN, ANN and CNN) make several important findings. Figure 8 provides a graphical comparison of the number of anomalies that each model identifies, thus, providing information that cannot be easily based on the scores of the performance. It can be seen that the proposed model achieves an optimal rate of anomaly detection without over- and under-detection like in the SVM/KNN models and deep-learn models respectively. Compared to traditional models like SVM, KNN and ANN, the proposed model performs better with a high F1 score (0.97) and a higher accuracy (0.94) and recall (0.99) which gives it a balanced result between recall and accuracy. The ROC-AUC score (0.50) a bit less, but that is actually an effective way of isolating anomalies with a high recall, which is important in anomaly detection when error of commission is expensive. KNN and SVM have more outliers (37), compared to the proposed model (30). This signifies that these models are excessively lenient when classifying outliers to be anomalies and hence chances of false positive are high. Such fewer outliers identified with the proposed model can be an indication of a more precise discrimination of actual anomalies and minimize the amount of noise over-detection. The accuracy (0.93) and perfect recall (1.00) of the neural-network models (RNN, ANN, CNN) is high but the models identify zero anomalies in the dataset. It shows that it could overfit to normal data and not be sensitive enough to outliers, which is a drawback that would lead to poor performance on anomaly detection in real-life settings. They also have low ROC-AUC of 0.50 which also indicates that their performance is similar to guessing randomly in the process of separate normal and anomalous points. The main advantage of the suggested framework is to optimise the rate of contamination of the Isolation Forest adaptively. This is possible, which means that the model will adapt more to developing data patterns. An RL agent is has a precision of 0.94 and high recall, which maximizes the contamination rate. This flexibility limits the false positives relative to contamination rate fixed models. There is variability observed in anomalies identified amongst models. The suggested model balances the anomaly count detected with the false alarms minimized better, but nevertheless, compared to SVM and KNN, the recommended model finds 30 anomalies, and it could be an indication of parameters favouring the aggressive detection approach. But the trade off of this approach is precision which is apparent in other dimensions by the poor performance measures of SVM and KNN. The experimental assessment is focused on showing the effectiveness of the adaptive tuning that is guided by reinforcement-learning in the offline mode. Though the extension of the assessment can be done using ablation experiments, computational overhead tests and controlled non-stationary data tests, they are not the subject of the present study and can be considered as important paths towards future research.

5 Conclusion

This paper suggests a scalable and dynamic process of effective anomaly detection in solar-generated power systems by means of incorporating reinforcement learning with the Isolation Forest algorithm. The new approach overcomes the major weaknesses of the traditional methods of negligence detection which require predetermined thresholds and manual optimization by dynamically updating the rate of contamination based on changes in the data distribution. This tuning is automatically done by the RL agent and keeps on learning new data thus making automatic and significant improvement to the long-term adaptability and performance. Experimental impressions illustrate a better outcome of RL-enhanced model, with 0.94 accuracy, 0.99 recall, and 0.97 F1 score- better performance as compared to the classic machine-learning models. High degree of accuracy is key in the energy systems, as irregularities that are not detected or identified properly may cause inefficiencies, threats to safety or unjustified maintenance expenses. In addition, the ability of the framework to adapt to noisy and drifting data makes it highly applicable in the implementation of the framework in actual renewable-energy conditions, where sensor reliability and weather variations are significant challenges.The envisaged RL-based anomaly detection system is a major step forward in smart energy monitoring. It fuses automation, flexibility, and high precision, creating a new benchmark for how modern renewable energy systems can be optimized and serviced in an increasingly dynamic and data-rich world. Its strength and autonomy make it particularly well suited to large-scale, distributed solar installations where centralized management may be constrained. Future work will include a comprehensive runtime and scalability evaluation of the proposed framework across larger datasets and heterogeneous computing environments, including multi-core and distributed platforms. Future work also Incorporating temporal sequence modeling via LSTM may improve the capability to identify time-dependent patterns. The versatility of the framework also means that it can be adapted to incorporate multimodal data from diverse sensors, which would allow for a more comprehensive understanding of system health. The experimental evaluation will be extended to include multiple randomized runs and cross-validation settings, enabling the reporting of quantitative results with error bars and focus on streaming and real-time environments, including latency analysis and online evaluation under practical deployment constraints.The design also lends itself well to deployment at the edge of computing, which would enable real-time processing and decision-making at the locations of energy generation—reducing latency and enhancing system resilience.

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Data Availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical Approval Not applicable.

Research Involving Human and /or Animals Not applicable.

Informed Consent Not applicable.

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